Geophys. J. Int. (1994) **116,** 131–140

New palaeomagnetic constraints on the position of central Bohemia during Early Ordovician times

J. Tait, V. Bachtadse and H. Soffel

Institut für Allgemeine und Angewandte Geophysik, Ludwig-Maximilians Universität, Theresienstr. 41, *München, Germany*

Accepted 1993 June 24. Received 1993 March 15; in original form 1992 September 18

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SUMMARY

Apparently conflicting data within the palaeomagnetic database suggest that Bohemia was at much shallower palaeolatitudes than the Armorican and Iberian massifs in Ordovician times. In order to resolve this controversy a palaeomagnetic study of Arenig and Llanvirn volcanic and sedimentary rocks from the Barrandian Basin, Central Bohemia has been carried out. This basin comprises a sequence of unmetamorphosed Lower Palaeozoic rocks, the main folding of which occurred in Late Devonian to Early Carboniferous times. The samples collected (17 sites, 153 samples) were subjected to stepwise thermal and alternating field treatment. Eleven sites yielded reliable magnetic directions, with analysis of the results using standard principal component and great circle analyses. Up to three directions of magnetization can be identified in these rocks. The first, termed A, is generally removed below 200 °C, although in some cases it persists up to 450 °C in the volcanic rocks. It is approximately parallel to the present-day Earth's magnetic field in the study area (Dec/Inc 360°/67°) and is thus thought to be of recent origin. The second direction identified, termed B, is isolated at intermediate blocking temperatures (150–350 °C), and yields an overall *in situ* mean direction of 195°/8°, $k = 22.3$, $\alpha_{95} = 13.1°$ (seven sites). These directions fail the fold test of McFadden (1990), and correspond to a palaeopole position of 34°S; 356°E. This coincides with the Late Carboniferous sector of the European apparent polar wander path, and thus the B direction is interpreted as being a secondary overprint of this age. The highest blocking temperature direction, termed C, is identified at temperatures between 350° and 450–600 °C. It passes the fold test, resulting in an overall mean direction of 312°/83°, $k = 21.9$, $\alpha_{95} = 14.6°$ (six sites) after bedding correction. This direction is interpreted as being representative of the Early Ordovician palaeomagnetic field direction in the Bohemian Massif and yields a palaeopole position of 58°N; 355°E. This translates into palaeolatitudes of 76°S for Bohemia, thus demonstrating that the Bohemian Massif was at similar peri-polar latitudes to the rest of Armorica during Early Mid-Ordovician times, and formed part of the northern margin of Gondwana.

Key words: Europe, palaeomagnetism.

INTRODUCTION

The hypothesis of Armorica as an independent plate during the Early Palaeozoic was introduced by Van der Voo (1979) on the basis of two apparently contradictory palaeomagnetic data sets. It was recognized that whilst the Late Precambrian–Early Cambrian segment of the apparent polar wander (APW) path from the Armorican Massif coincided with that of Gondwana, by Late Devonian times the palaeomagnetic data suggested that the Armorican Massif was contiguous with northern Europe (Van der Voo 1979, 1982, 1988; Perroud, Van der Voo & Bonhommet 1984), whilst the northern margin of Gondwana was at higher palaeolatitudes (Perroud *et al.* 1984). Thus, the term Armorica was adopted to refer to the various Hercynian massifs now contained within the Hercynian and Alpine fold belts of Europe. However, as noted by Van der Voo (1988), the palaeomagnetic data from southern Britain (which he

included as part of Armorica) indicated shallower palaeolatitudes in Mid and Late Ordovician times than those from the Armorican and Iberian massifs. In the palaeontological data, a faunal distinction between southern Britain and the Hercynian massifs of continental Europe in Late Ordovician times has been recognized for some time (Cocks & Fortey 1982), suggesting that these two areas were not contiguous throughout the Ordovician. Recent palaeomagnetic data from England and Wales have supported this hypothesis, and indicate that in Early Ordovician times southern Britain rifted from the margin of Gondwana and drifted northwards as part of an independent plate, Avalonia, whilst the rest of 'Armorica' (*sensu* Van der Voo 1988) retained its African affinities. Avalonia extends from coastal New England through Nova Scotia and the Avalon Peninsula of Newfoundland to SE Ireland, Wales, England and the Ardennes. The remainder of Hercynian Europe, i.e. the Armorican, Iberian and Bohemian massifs are what are referred to as Armorica throughout this paper.

It has been suggested that Avalonia should be subdivided into Eastern and Western Avalonia (Perroud *et al.* 1984; Van der Voo 1993), based on palaeomagnetic data from the Piedmont metamorphic rocks in northern Delaware by Brown & Van der Voo (1983). However, these palaeomagnetic data are of fairly poor quality with no field tests with which to constrain the age of magnetization. Furthermore, given the possibility of internal rotations, recent palaeomagnetic data from rocks with a whole rock Rb–Sr age of 434 ± 15 Myr from the Dunn Point, Nova Scotia by Johnson & Van der Voo (1990), yield a palaeolatitude for western Avalonia (41°S) which is in good agreement with Late Ordovician data from southern Britain. In addition, there is no faunal evidence for any oceanic separation of these two domains at any time during the Ordovician (McKerrow, Dewey & Scotese 1991). Thus, in the reconstruction given in this paper, Avalonia is considered to have remained a coherent tectonic element during its northward drift from Gondwana and subsequent suturing with Baltica in Siluro-Ordovician times.

From palaeomagnetic evidence it is clear that the northern margin of Gondwana was located at peri-polar latitudes during the Early Ordovician (Van der Voo 1993, and references therein), and evidence that Armorica originated from this margin and was peripheral to Gondwana during the Early Ordovician is strong. It is indicated in the structural record by recognition of the Cadomian unconformity in the Armorican Massif (Rutten 1969, and references therein), the presence of Hirnantian glaciomarine deposits in both Iberia and the Bohemian Massif (Brenchley *et al.* 1991), and also in the faunal record. The characteristic high latitude 'Mediterranean' or 'Bohemian' fauna persisted over the cratonic areas of northern Gondwana and Armorica and remained distinct from Baltic and Laurentian faunas during the Llandeilo and most of the Caradocian (Spjeldnaes 1967; Fortey & Cocks 1988).

These faunal differences between the northern margin of Gondwana, which included Avalonia and Armorica during Early Ordovician times, and Baltica led Cocks & Fortey (1982) to propose the existence of the Tornquist Sea separating the two domains, a model which has been further supported by subsequent palaeontological studies (Fortey & Cocks 1988; Young 1990; Cocks & Fortey 1990; Harper 1992; Parkes 1992). This model is also supported by palaeomagnetic data, as reviewed by Bachtadse *et al.* (1993), which shows that Baltica drifted away from Gondwana into subtropical latitudes during Mid–Late Ordovician times, thus widening the Tornquist Sea (Torsvik *et al.* 1992). The opening of the Rheic Ocean between Gondwana and Avalonia caused Avalonia to migrate northwards, and by Late Ordovician times it was located in temperate palaeolatitudes (McCabe, Channell & Woodcock 1992).

An alternative model to this has been proposed by Paris & Robardet (1990). They base their arguments on planktonic (acritarch/chitinozoa) fossils to refute the presence of the Tornquist Sea between Avalonia and Baltica, and call the ocean separating Gondwana (including Armorica) and Baltica (including Avalonia) the Rheic Ocean. However, it has been demonstrated by Cooper, Fortey & Lindholm (1991) that planktonic fauna may be pandemic, even during the period of high provinciality that was evident during the Early Ordovician. In their recent analysis of the reliability of different fossil groups with regards to determining palaeogeographic boundaries, Fortey & Mellish (1992) and Fortey & Cocks (1992) have demonstrated that the lack of the Tornquist Sea in the model of Paris & Robardet is probably an artefact of their having used planktonic fossils rather than the more endemic benthic fauna such as trilobites, ostracods and brachiopods.

Close affinities between Armorica and Gondwana during Ordovician times are also indicated by palaeomagnetic data (Perroud *et al.* 1984). Results from the Ibero-Armorican Arc in western France and Spain place it at high southerly palaeolatitudes in Early Ordovician times (Duff 1979; Perroud & Bonhommet 1981; Perroud, Bonhommet & Thébault 1986), and data from Upper Ordovician rocks indicate that the Armorican Massif remained adjacent to the northern margin of Gondwana (Perroud & Van der Voo 1985), in agreement with the lithological and faunal indicators described above. In contrast to this, however, Perroud, Bonhommet & Van der Voo (1983) reported results from Upper Ordovician dolerites of the Crozon Peninsula, France. Their bedding-corrected 'C' direction of magnetization gives palaeolatitudes of 46°S, suggesting that Armorica had already rifted from Gondwana and was moving towards the northern continents. However, the pole position derived from their C direction plots close to the Permo-Carboniferous sector of the European APW path. In the absence of a conclusive fold test, the Crozon palaeopole position may, therefore, represent a younger magnetization age as discussed by the original authors (Perroud *et al.* 1983).

The Bohemian Massif is thought to constitute the eastern extremity of Armorica, and correlations with the Armorican Massif on a geological basis have been made (Tollmann 1982). However, this cannot be quantitatively demonstrated due to the sparse and ambiguous palaeomagnetic data from the Bohemian Massif for Early Palaeozoic times. Andreeva, Bucha & Petrova (1968) reported steep (>60°) inclinations from Bohemia, which, although they do not stand up to modern reliability criteria, nevertheless agree with Early Ordovician data from the western margin of the Bohemian massif (Böhm 1989; Tait *et al.* 1993). Early Ordovician results by Krs *et al.* (1986, 1987) on the other hand, indicate

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much shallower inclinations corresponding to palaeolatitudes of 40°S. Although this is in apparent contradiction to lithological indicators, such as Hirnantian glaciomarine deposits (Storch 1990; Brenchley *et al.* 1991), and most of the data from the Armorican massif (Duff; Perroud & Bonhommet 1981; Perroud *et al.* 1986), it is similar to the data of Perroud *et al.* (1983) from the Crozon dolerites in northern France. However, it has been suspected for some time that the directions obtained by Krs *et al.* (1986, 1987) are not true primary directions but represent secondary overprints of a younger, possibly Devonian–Carboniferous age (Piper 1987; Van der Voo 1993).

The present study was undertaken with the aim of resolving some of these discrepancies with regard to the Bohemian Massif. New palaeomagnetic data from Early Ordovician rocks of the Barrandian Basin in the Bohemian massif are presented and the resulting palaeogeographic reconstructions for Early Ordovician times are discussed.

GEOLOGICAL SETTING AND SAMPLING

The Barrandian Basin is part of the Tepla–Barrandian Terrane which constitutes a subunit within the Moldanubian Zone of the Hercynian belt in Europe. It comprises a synclinal structure within the gently folded lower greenschist facies rocks of the Cadomian basement, overstepped by a sequence of unmetamorphosed Lower–Mid Palaeozoic rocks called the Barrandian Sequence (Franke 1990). The geographic location of this basin and a sketch map of the geology is given in Fig. 1.

This sequence comprises Lower Cambrian fluviatile conglomerates, overlain by Ordovician clastic rocks of low diversity. They consist predominantly of shales, mudstones and quartzites with evidence of minor transgressive and regressive cycles, culminating in the widely recognized Upper Hirnantian transgressive epidosde which is related to eustatic sea level changes (Brenchley *et al.* 1991).

Middle and Upper Ordovician deposits of the Barrandian Basin are characterized by a monotonous sequence of mudstones and deep water shelf facies, with two thin intervals of Hirnantian age glaciomarine deposits (Brenchley & Storch 1989). These in turn are overlain by Silurian and Devonian, up to Givetian, rocks which comprise predominantly shales and carbonates, with the first carbonate rocks appearing in Mid-Silurian times (Havlíček 1981).

The most important phase of volcanic activity during the Ordovician of the Barrandian Basin was initiated in the Tremadoc under shallow marine conditions, and lasted almost uninterrupted until Early to Mid-Arenig times. This resulted in the basic explosive–effusive Komárov complex which is up to 1000 m in thickness. This complex was formed by fissure eruptions along a trend which ran at an angle to the main axis of the basin, reviving pre-Ordovician Cadomian structures (Havlíček 1981).

Samples from Upper Arenig and Lower Llanvirn sediments and interbedded volcanic rocks in the Barrandian Basin were collected, the localities of which are shown in Fig. 1, and the lithologies are listed in Table 1. Since the pioneering work of Barrande in the 19th century, these fossiliferous sediments have been extensively studied and the biostratigraphic ages are well established. The structure of these Lower Palaeozoic rocks is fairly simple, with the Lower Cambrian to Givetian rocks forming a gently dipping syncline, unconformably overlain by Westphalian age sediments. From these relationships and comparison with surrounding areas in the Tepla–Barrandian, where deformation was initiated in Late Givetian times with a thermal event dated at 380 Myr from K–Ar studies (Müller-Sohnius *et al.* 1987), deformation in the Barrandian Basin is thought to be Late Devonian–Early Carboniferous in age (Franke 1990). There is also evidence of syn-sedimentary deformation within the Barrandian Basin. Studies of the thickness and facies development by Havlíček (1981) have shown that much of the faulting and related folding resulted from a

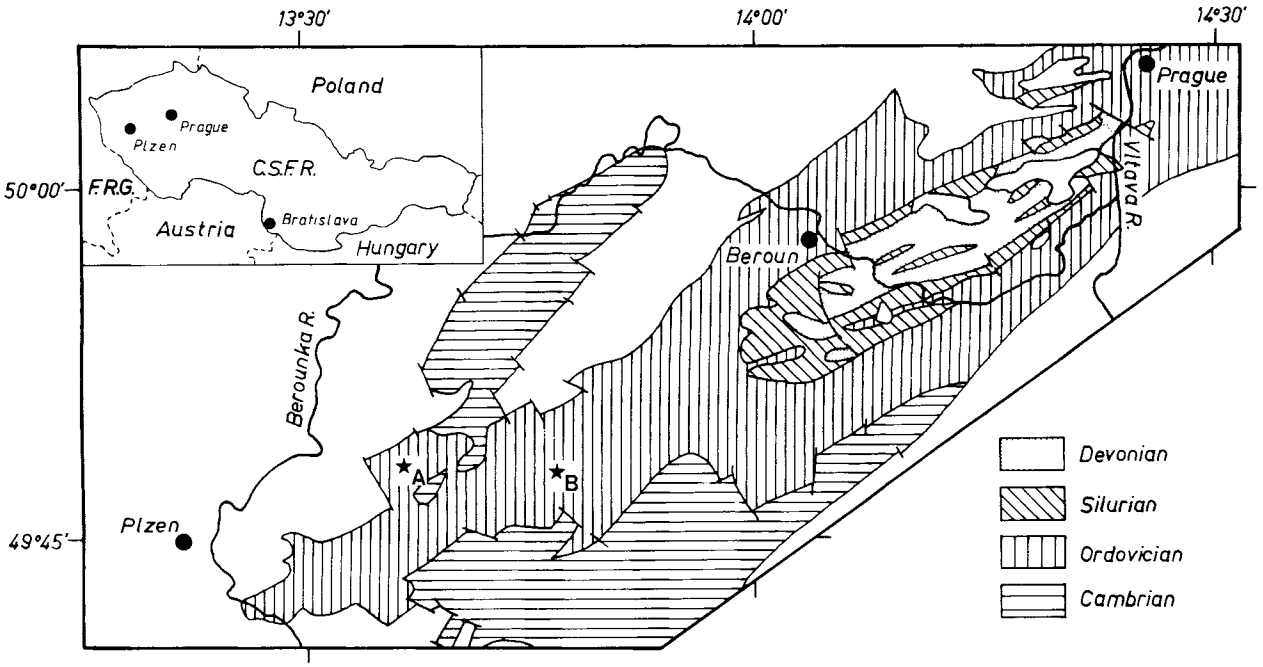


Figure 1. Geological sketch map of the sampling areas within the Barrandian Basin and its location within eastern Europe.

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Table 1. Site mean and overall mean directions for components B and C. N indicates the number of samples measured, *n* the number of sample directions used in the calculations, and the number in brackets indicates the number of great circles. Also listed are the values for the precision parameter *k*, and the half angle for the cone of 95 per cent confidence. Aren. indicates an Arenig age, and Llan. indicates a Llanvirn age; sandst. indicates sandstone, and p. lava pillow lavas.

			Component B			
				in situ		
site	lithology	age	N/n	Dec/Inc	k	α_{95}
EJP2	oolite	Aren.	5/5	212/+21	50.3	10.9
EJP3	oolite	Aren.	5/3	213/+16	44.2	18.8
EJP1	tuffite	Llan.	10/10(8)	206/+1	54.3	7.3
EJP4	sandst.	Llan.	10/0	-/-	-	-
MYT1	shales	Llan.	3/3	192/-2	43.3	19.1
MYT2	p.lava	Llan.	7/0	-/-	-	-
MYT3	p.lava	Llan.	5/5	192/+18	15.1	20.3
MYT5	p.lava	Llan.	7/0	-/-	-	-
MYT6	tufficite	Llan.	4/2	180/+7	-	-
MYT7	p.lava	Llan.	10/0	-/-	-	-
MYT8	shales	Llan.	6/4	178/-1	25.5	18.6
mean			7 sites	195/8	22.3	13.1
palaeopole				34°S; 356°E		

	Component C						
		in situ			*bedding corrected*		
site	N/n	Dec/Inc	k	α_{95}	Dec/Inc	k	α_{95}
EJP2	5/0	-/-	-	-	-/-	-	-
EJP3	4/0	-/-	-	-	-/-	-	-
EJP1	10/8(8)	049/+51	35.4	10.9	033/+82	35.4	10.9
EJP4	10/3	252/+67	13.1	35.2	293/+76	109.3	12.1
MYT1	3/0	-/-	-	-	-/-	-	-
MYT2	7/7	227/58	19.2	14.1	256/66	19.2	14.1
MYT3	5/0	-/-	-	-	-/-	-	-
MYT5	7/5(2)	209/+72	19.7	21.6	278/+81	19.7	21.6
MYT6	4/0	-/-	-	-	-/-	-	-
MYT7	10/7(2)	242/+78	10.3	20.6	332/+71	22.5	13.2
MYT8	6/2	080/+84	-	-	086/+68	-	-
mean	6 sites	230/+80	10.2	22.0	312/+83	21.9	14.6
palaeopole		---			58°N; 355°E		

long, essentially syn-sedimentary, process throughout Ordovician and Silurian times, culminating after the close of sedimentation in the Givetian.

The samples collected in this study are from beds of uniform orientation within site, but intersite variation of the beds allows for a fold test on a regional level. The structural attitude of the volcanic horizons was determined from immediately overlying sediments in all cases. The samples were collected using a petrol-driven rock drill, with orientation by means of a standard sun compass and/or Brunton magnetic compass.

LABORATORY PROCEDURES

The field-drilled cores were cut into standard 2.2 × 2.5 cm specimens and magnetizations measured using a two axis 'Remanomètre à squid RS 01' (LETI, Grenoble) cryogenic magnetometer in Munich, or a 2G cryogenic magnetometer at ETH, Zurich. The samples were demagnetized using stepwise thermal, or alternating field (A.F.) techniques, and the magnetic susceptibility measured after each heating step on a KL2 Kappabridge in order to monitor mineralogical alteration.

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The demagnetization results were analysed using orthogonal vector plots (Zijderveld 1967) and stereographic equal angle projection. The linear (planar) demagnetization trajectories were identified by eye using at least three (four) successive data points, and the directions of the linear segments and poles to the planes were calculated using the principal component analysis of Kirschvink (1980). Where overlapping coercivity or blocking temperature (T_B) spectra prevented complete separation of the components, great circle analysis using the method of McFadden & McElhinny (1988) was carried out on the planar demagnetization trajectories to identify the underlying direction. This program uses an iterative procedure whereby the maximum likelihood estimate of the endpoint direction within the acceptable sector for each great circle can be determined. Where possible, this is then combined with stable endpoint directions, resulting in determination of the best estimate of the mean direction, using Fisher (1953) statistics in all cases. If no endpoints are available, then a mean direction can still be calculated with the use of sector constraints.

In samples containing three directions of magnetization, it is possible that the T_B of the lower and intermediate temperature directions have overlapping spectra at lower temperatures, whilst the intermediate and high temperature directions have overlapping spectra at higher temperatures. This results in the identification of two distinct great circles, but may prevent direct identification of both the intermediate and high temperature directions. In such cases, great circle analysis of both these directions can be carried out in two separate iterative procedures using the program of McFadden & McElhinny (1988).

PALAEOMAGNETIC RESULTS

The natural remanent magnetization (NRM) intensities varied from 0.2 to 29 mAm^{-1} for the sediments, and from 7 to 400 mAm^{-1} for the volcanic rocks, and most of the samples measured show multivectorial behaviour during thermal demagnetization. The sediments, in particular, tend to undergo mineralogical alteration during thermal treatment, and so A.F. demagnetization was attempted. However, this method failed, in most cases, to reveal the full complexity of the demagnetization trajectories, and so the results presented in this paper are based exclusively on thermal demagnetization treatment. Of the 17 sites studied, only 11 yielded reliable palaeomagnetic vectors. For the remaining six sites, which comprise red shales and andesites, the magnetizations were either dominated by a component parallel to the present-day local Earth's magnetic field, or there was no within-site consistency of results. Rock magnetic studies (isothermal remanent magnetization (IRM) and thermomagnetic experiments) of samples from sites yielding reliable palaeomagnetic vectors, indicate magnetite as the principal remanence carrier.

During thermal demagnetization the sediments tend to show fairly complex behaviour, and due to mineralogical alteration stable endpoints are only rarely reached. The oolites are characterized by removal of a soft component of magnetization between 20° and 200 °C. This component, termed the A direction, is in general parallel to the direction of the present-day Earth's magnetic field (360°/67°) *in situ*. It is thus considered to be of recent origin and will not be discussed further. Above this temperature, linear segments could not typically be identified in the results from the oolites. However, the directions from successive temperature steps between 300° and 450 °C group around a shallow southerly direction of magnetization. This direction of intermediate T_B is termed the B component.

The scenario for the tuffites is rather more complicated. After removal of A below 200 °C, and during removal of B between 240° and 440 °C, the demagnetization trajectory is not directed towards the origin, but is moving towards an easterly direction (Fig. 2a), clearly indicating the presence of a third, higher temperature direction which is called C. At temperatures higher than 440 °C, however, the magnetization directions become erratic due to mineralogical alteration and so the direction of C cannot be observed directly. Stereographic projection of results from these samples clearly demonstrates that the demagnetization trajectories follow two distinct great circle paths (Fig. 2b). For each great circle defined, the successive directions describe a single demagnetization plane showing that only two directions are being removed. The lower temperature path from 100° to 200 °C moves from a northerly direction towards the southerly and shallow B direction, and results from the removal of A and B. The second path between 240° and 440 °C moves away from B towards a much steeper, easterly direction, clearly indicating the presence of C, the direction of which cannot be isolated.

The results obtained from the volcanics and interbedded sediments are, in general, similar to those described above (Fig. 3). Most samples are characterized by the removal of the present-day A component which often persists up to 450 °C before removal of either B (shallow southerly direction) or C is observed. Only occasionally do samples contain all three directions of magnetization. For sample MYT801 (Fig. 3c) the demagnetization trajectory follows a great circle path from 200° to 400 °C with removal of an intermediate northerly direction of magnetization, to reveal C at higher temperatures. However, this northerly direction is not identified in any other samples and so no interpretation can be placed upon it. Stable endpoints for the C direction are usually obtained (Fig. 3), but occasionally the demagnetization trajectory continues to curve through the last measurement. In cases such as these, great circle analysis using the method of McFadden & McElhinny (1988) was carried out.

INTERPRETATION

In the tuffite samples from site EJP1, no direct observations of the B direction can be made. From the two great circles defined by each sample (Fig. 2b) and their intersection, however, it is clear that the B direction must be of a fairly shallow southerly direction. In order to determine the direction more precisely, great circle analysis using the lower temperature path, which moves from northerly (or steep westerly) to shallow southerly directions, was carried out as described above. Entering suitable sector constraints for the position of B on each of the 10 great circles identified, a best-fit *in situ* direction of 206°/1°, $k = 54.4$, $\alpha_{95} = 7.4°$, $n = 10$ (203°/−26°, $k = 54.4$, $\alpha_{95} = 7.4°$ after bedding correction) is obtained. These directions are in

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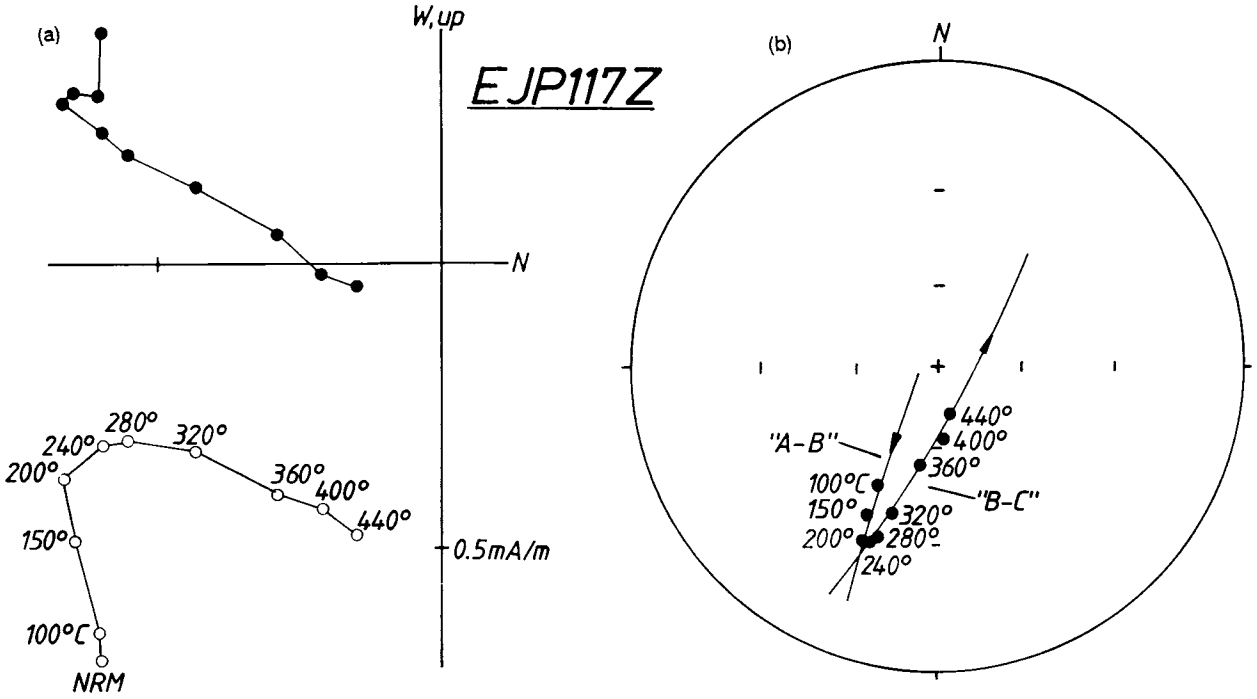


Figure 2. Orthogonal projection of thermal demagnetization characteristics (solid (open) symbols represent the horizontal (vertical) component), and (b) equal angle lower hemisphere stereographic projection of the tuffites given in geographic coordinates. In (b) two great circle paths can be clearly identified, the first from 100° to 200 °C resulting from removal of A and B, and the second from 240° to 440 °C resulting from removal of B and C. At temperatures higher than 440 °C the magnetization directions become random due to mineralogical alteration and no further reliable measurements could be made.

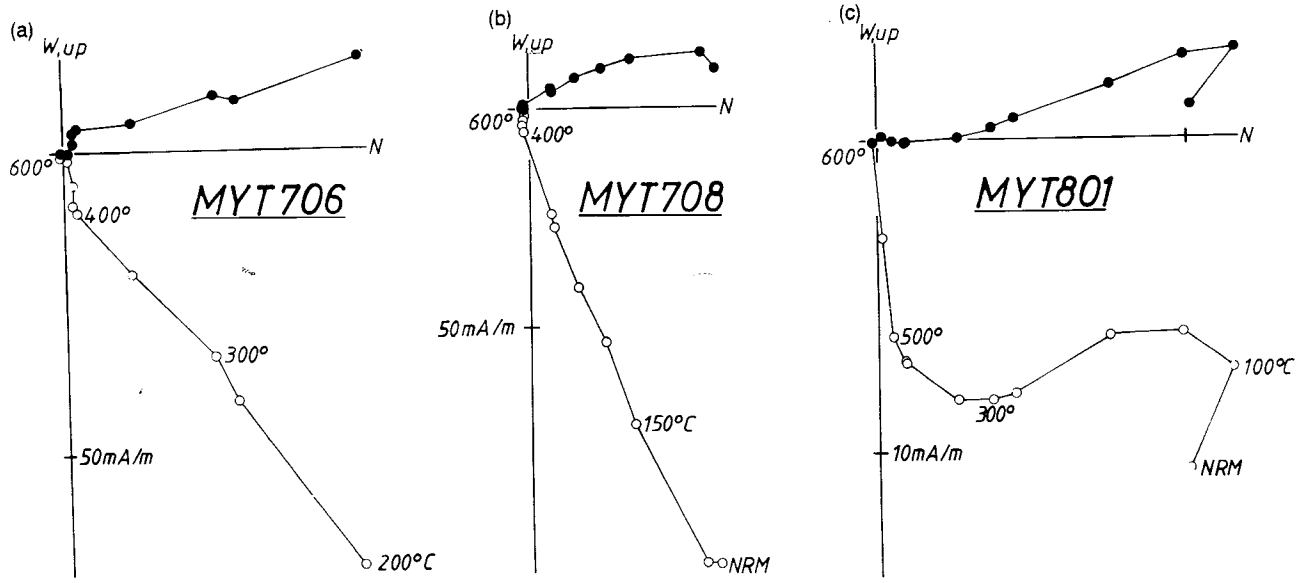


Figure 3. Thermal demagnetization characteristics of the volcanic (a and b) and associated sedimentary (c) rocks. (a), (b) show removal of an A direction of magnetization, to reveal the steeper C direction above 400 °C (a and b) and 500 °C (c).

fairly good agreement with the directions obtained from direct observation of the demagnetization trajectories in other samples in which straight line segments for B can be identified (Table 1, Fig. 4a).

Combining the seven site mean directions obtained for B (as listed in Table 1 and shown in Fig. 4b) gives an overall *in situ* mean direction of 195°/8°, $k = 22.3$, $\alpha_{95} = 13.1°$, and 198°/12°, $k = 5.9$, $\alpha_{95} = 24.9°$ after tilt correction. The directions fail the fold test of McFadden (1990), demonstrating that this component was acquired after structural deformation and represents a secondary overprint. Folding of these rocks occurred at latest in Early Carboniferous times during the Hercynian orogeny, by which time it is clear from geological and palaeomagnetic evidence (Van der Voo 1983; Perroud *et al.* 1984) that Armorica and the Bohemian Massif had collided with the

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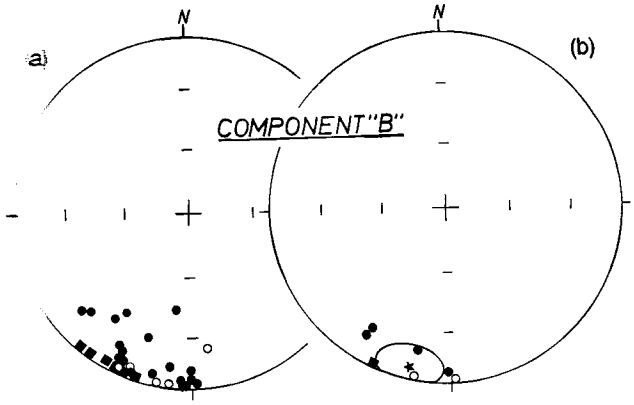


Figure 4. Equal area projection of the *in situ* sample (a) and site mean (b) directions for B. Solid (open) symbols represent lower (upper) hemisphere, square symbols represent directions obtained from great circle analysis. The star in (b) indicates the overall mean direction with its cone of 95 per cent confidence.

northern continents and constituted part of the Old Red Continent. When the pole of 34°S; 356°E derived from B is compared with the APW path for Europe (Van der Voo 1993), it corresponds to the Late Carboniferous sector and is therefore considered to result from a remagnetization event of this age. Secondary magnetization directions of this age have been recognized in pre-Carboniferous rocks throughout the Bohemian Massif, and indeed throughout Europe. Whether this is related to widespread fluid migration (see McCabe & Elmore 1989) and the formation of secondary magnetic minerals during the Hercynian orogeny, or is the result of other physical processes is beyond the scope of this paper.

In those samples where C could not be isolated and great circles are defined at intermediate to high temperatures, great circle analysis was carried out. For the tuffites from sites EJP1 no direct observations of the C direction could be made. However, it is clear that the endpoint must lie somewhere on the great circle after the last measurement (Fig. 2b). Entering suitable sector constraints into the

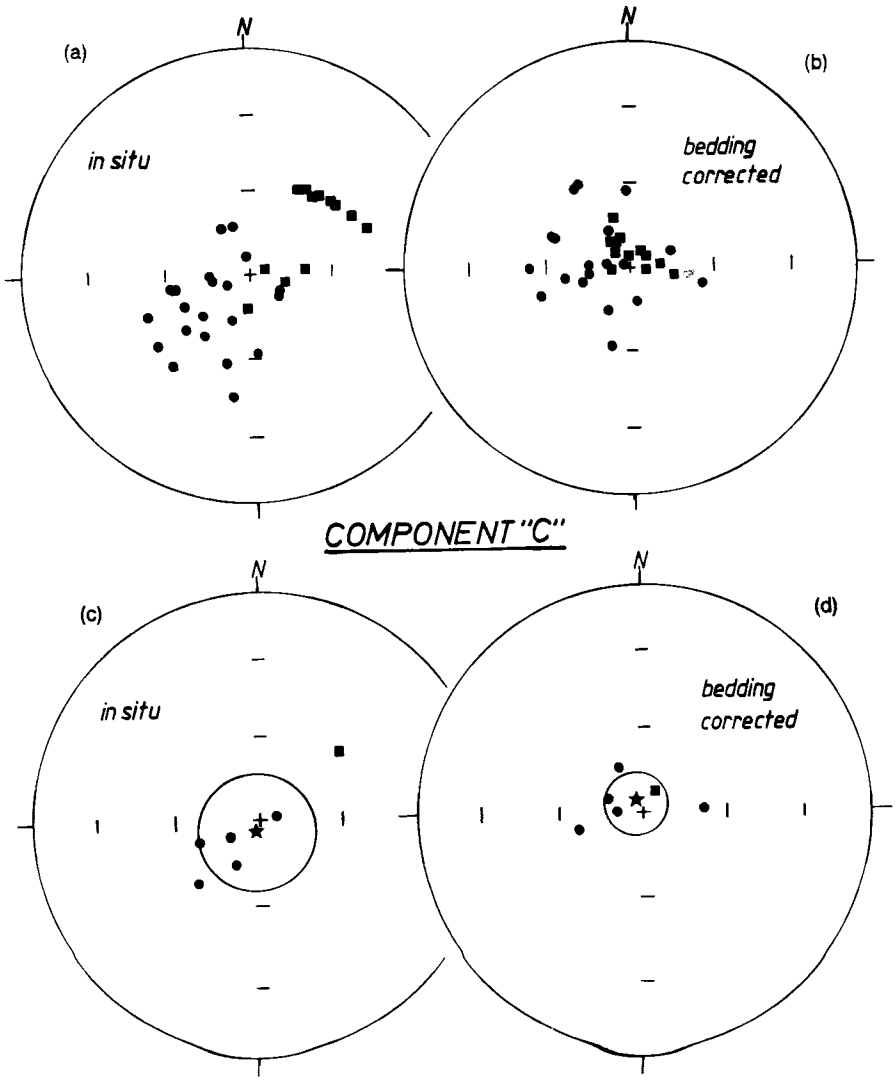


Figure 5. Equal area projection of the *in situ* (a) and bedding-corrected (b) sample directions for C, and the *in situ* and bedding-corrected site mean directions (c and d, respectively). Notation as for Fig. 4. Also shown are the overall mean directions for C (star) and cones of 95 per cent confidence both before and after bedding correction.

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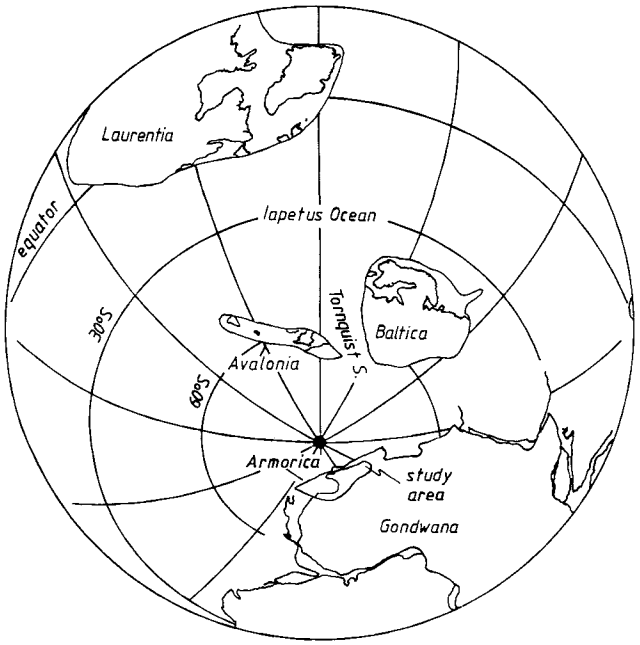


Figure 6. Palaeogeographic reconstruction of the Iapetus bordering continents during Llanvirn times with the position of Armorica according to data from the Barrandian Basin (this study). The positions of Laurentia and Gondwana are positioned according to Van der Voo (1993), Avalonia according to Torsvik *et al.* (1993), and Baltica according to Torsvik *et al.* (1992).

procedure of McFadden & McElhinny (1988), a best-fit *in situ* solution of 049°/51°, $k = 35.4$, $\alpha_{95} = 10.9°$, $n = 8$ (033°/82°, $k = 35.4$, $\alpha_{95} = 10.9°$ after bedding correction) is obtained.

Direction C is identified with single polarity (Figs. 5a and b) and the six site mean directions (Table 1) combine to yield an overall mean direction of 230°/80°, $k = 10.2$, $\alpha_{95} = 22.0°$ *in situ*, and 312°/83°, $k = 21.9$, $\alpha_{95} = 14.6°$ after bedding correction (Fig. 5c and d). The statistical distribution of the sample mean directions was tested during stepwise unfolding of the rocks using the procedure of McFadden (1990). A peak in the values was obtained after 100 per cent unfolding and the age of magnetization is thus interpreted as being closely bracketed with the rock age, which for the volcanic rocks is considered to be Llanvirn from stratigraphic relationships.

Assuming a reversed polarity for the bedding-corrected C direction, the resulting palaeo (south) pole position of 58°N; 355°E translates into palaeolatitudes of 76°S for the central Bohemian Massif during Early Ordovician times. The palaeogeographic reconstruction based on this pole position for Armorica is shown in Fig. 6. From this it is evident that Armorica was positioned on the northern margin of Gondwana at this time.

Although the rocks were not sampled for magnetostratigraphic purposes, the presence of exclusively reversed directions is in basic agreement with the magnetic polarity pattern proposed by Torsvik & Trench (1991) for the Early Ordovician.

CONCLUSIONS

Data from Lower Ordovician rocks elsewhere in Armorica are in agreement with the results obtained in this study (Duff 1979; Perroud & Bonhommet 1981; Perroud *et al.* 1986). Similarly, the data by Perroud & Van der Voo (1985) from France, and by Böhm (1989) and Tait *et al.* (1993) from the western margin of the Bohemian Massif from Upper Ordovician rocks also indicate that Armorica remained at high southerly latitudes throughout the Ordovician. However, earlier results from the Barrandian Basin by Krs *et al.* (1986, 1987) are in marked contrast to the present study. They yield much shallower inclination values, corresponding to palaeolatitudes of 40°S.

The results of Krs *et al.* (1986) are based on tuffites from the same locality as the present study, and also from red silicites of similar age. The tuffite samples also follow a great circle path (see fig. 6 of Krs *et al.* 1986), but, although they are not subjected to full vector analysis treatment, the demagnetization trajectories appear to be moving towards a direction with steep negative inclination values *in situ*, the significance of which is not clear. The silicite samples reveal thermally stable directions of magnetization, but the direction yielded is thought to represent either a younger magnetic overprint, or possibly is the result of superimposed directions of magnetization.

Thus, the new palaeomagnetic data clearly demonstrate that the Bohemian Massif was at similar high palaeolatitudes to other areas within Armorica in Early Ordovician times, and adjacent to the northern margin of Gondwana. This is in agreement with faunal indicators, i.e. the predominance of the 'Bohemian' fauna in both Armorica and northern Gondwana until the Caradoc (Spjeldnaes 1967; Fortey & Cocks 1988; Cocks & Fortey 1990).

Within the overall context of the palaeogeographic evolution of the Iapetus bordering continents during Early Palaeozoic times, it is clear that whilst Gondwana remained at high peri-polar latitudes and Laurentia remained at equatorial latitudes during the Ordovician (Van der Voo 1993), Baltica drifted rapidly northwards from high southerly latitudes to occupy an equatorial position by the Late Ordovician (Torsvik *et al.* 1992). Fragmentation of the northern margin of Gondwana was also initiated in Early Ordovician times as Avalonia began to drift northwards (Cocks & Fortey 1990; McKerrow *et al.* 1991; Channell *et al.* 1992), whilst Armorica remained peripheral to Gondwana.

ACKNOWLEDGMENTS

We are particularly grateful to M. Krs and P. Pruner for their help in arranging fieldwork in the Barrandian Basin, and their invaluable assistance in conjunction with O. Fatka with the local geology. We would also like to thank two anonymous reviewers for their helpful comments from which the manuscript benefitted. Many thanks are due also to W. Lowrie for granting us access to the palaeomagnetic laboratory at the ETH Zurich, where some of the measurements were carried out, and our thanks also go to F. Heller (ETH Zurich) for his strong support. Thanks are due to F. Maier for his help in collecting the samples, and to A. C. Choukèr and A. Zitzelsberger for help with some of the rock magnetic measurements. This work was supported by grant Fr668/6-2 from the German Research Council (DFG). VB also acknowledges financial support by SFB69, Berlin.

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